

MICRO 3 — Advanced Microeconomics · Revision Sheet

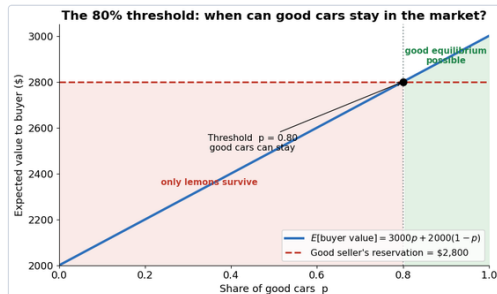
Lecture-by-lecture, topic-by-topic · concept → method/when-to-use → worked example · all problem-set question types

LECTURE 1 — ASYMMETRIC INFORMATION

1.1 Adverse Selection & the Lemons Market

Hidden **quality**, known to the seller, not the buyer (**pre-contract**). Buyers pay an average price → best types exit → average quality & price fall → **death spiral** (thinning → collapse).

► **Method / use when** Find the **Rational-Expectations Equilibrium (REE)**: (1) assume ALL types trade; (2) $E[V] = \sum p_k V_k$; (3) if $E[V] <$ the top type's reservation, that type exits; (4) recompute on the smaller pool; (5) stop at the largest self-consistent set. Trigger: "buyers can't tell quality", "used cars", "which types are traded".



Good eq. exists only if $E[V]$ clears the good-seller reservation (here $p \geq 0.8$); lemons eq. always co-exists.

► **Example.** 3 types, equal numbers; values (seller, buyer): Good(1000, 1500), Med(500, 750), Bad(100, 150). All: $E[V] = \frac{1}{3}(1500 + 750 + 150) = 800 < 1000$ → Good exit. Med+Bad: $E[V] = \frac{1}{2}(750 + 150) = 450 < 500$ → Med exit. Bad only: $150 \geq 100$ ✓. **Price 150, only lemons trade.**

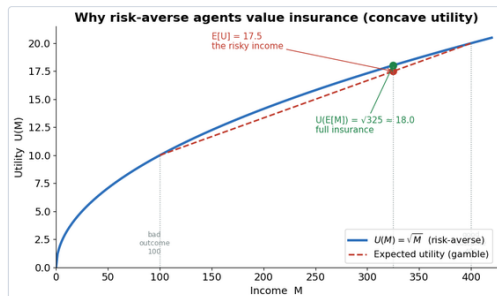
► **Example.** Partial unravelling: 5 Good, 20 Med, 5 Bad. Good exit ($E[V] = 775 < 1000$), but Med+Bad pool is now 80% Med: $E[V] = \frac{4}{5}(750) + \frac{1}{5}(150) = 630 \geq 500$ → Med & Bad trade. Concentration of mid-quality saves the market.

1.2 Risk Aversion & Insurance

Concave utility $U = \sqrt{M} \Rightarrow E[U(M)] < U(E[M])$: a risk-averse agent prefers a sure thing → will pay to remove risk.

Actuarially fair premium = $E[\text{insurer payout}]$ (zero profit pool). Risk-based premiums: each type pays its own expected cost.

► **Method / use when** Uniform premium → healthy opt out, pool gets sicker, price rises (adverse selection). Test each candidate pool for self-consistency (premium = avg cost of those who stay, & each stayer's RP ≥ premium). Trigger: "single premium", "who buys", "government subsidy".



Concave U: utility of certain income > expected utility of the gamble → pay a fair premium.

► **Example.** Costs/RP: H 1000/2000 (25%), M 2000/3000 (50%), L 4000/6000 (25%). Uniform & all buy: $\text{avg} = 2250 >$ H's RP 2000 → H out. M+L: $\text{avg} = 2000/0.75 \approx 2667$, both RPs exceed it ✓. 25% uninsured. Subsidy to make H join = $2250 - 2000 = 250/H$ (also drops M, L price to 2250).

1.3 Moral Hazard

Hidden **action**, **post-contract**. Once insured, the price of care → 0, so the insured **over-consumes** (drives recklessly, over-treats).

► **Method / use when** **Uninsured**: pay true price $P=MC$, quantity off own demand. **Fully insured**: $P=0$ → moral-hazard quantity; the fair premium is computed on these **larger** quantities, so it exceeds uninsured spend. **DWL** = units whose value < MC. Trigger: "after buying insurance", "expected medical spend".

► **Example.** $\text{Pr}(\text{mild, mod, sev}) = \frac{1}{2}, \frac{1}{3}, \frac{1}{6}$; demands $5 - P, 15 - P, 20$; $MC = 2$. Uninsured ($P=2$): $Q = 3, 13, 20$ → spend = $\frac{1}{2}(6) + \frac{1}{3}(26) + \frac{1}{6}(40) = 18.33$. Insured ($P=0$): $Q = 5, 15, 20$ → fair premium = $21.67 > 18.33$; gap = moral-hazard loss.

► **Example.** Agency (physician vs HMO): Treatment A cost 1200, $E[U] = 2$; B cost 2400, $E[U] = 15$. Physician (ignores cost) picks B (max utility). HMO (utility/\$): B gives $15/2400 >$ A's $2/1200$ → also B, but the conflict of objectives is the moral-hazard/agency point.

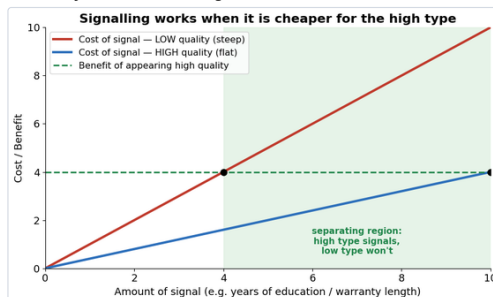
1.4 Signaling & Screening

Signaling = informed party takes a costly action to prove quality. Works only under **single-crossing**:

► **Method / use when**

Cost(low) > gain from being believed high > Cost(high)

⇒ **separating eq.**: high types signal, low don't. **Screening** = the uninformed offers a menu so types self-select. Trigger: "warranty", "education/degree", "menu of contracts".



Single-crossing: signal cheap for the high type, dear for the low type.

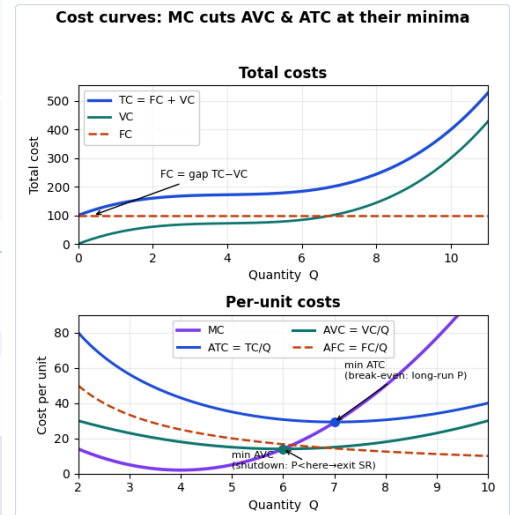
► **Example.** Warranty costs good seller 300 but lemon seller 1300. Only good cars carry it → warranty = credible quality signal. Spence: a degree can separate workers even with zero productivity gain (socially wasteful).

LECTURE 2 — EQUILIBRIUM IN DIFFERENT MARKET STRUCTURES

2.1 Costs & cost curves

Economic Π counts **implicit/opportunity** cost, so < accounting Π . **Sunk costs are ignored**. $TC = FC + VC$; $MC = dTC/dQ$; $AFC = FC/Q$; $AVC = VC/Q$; $ATC = AVC + AFC$.

► **Method / use when** **MC cuts AVC & ATC at their minima**. Decisions use only avoidable/marginal costs. Trigger: "should they accept/continue?", "opportunity cost", "derive AVC/ATC/MC".



Totals ($TC = FC + VC$) & per-unit; MC passes through min AVC (shutdown) and min ATC (break-even).

► **Example.** Plane already flying (cost sunk); extra passenger $MC = 0$; offer $300 > 0$ → **accept** (ignore the \$500 avg cost). $TC = 5 + 40Q + 20Q^2$ → $AVC = 40 + 20Q$, $MC = 40 + 40Q$, ATC min at $Q = 0.5$ ($= 60$), where $MC = ATC$.

► **Example.** **Opportunity cost**: degree costs 35k tuition + 3k books + 40k forgone wages = $78k/yr \times 3 = 234k$. Benefit $5.5k \times 40yr = 220k < 234k$ → **don't study** (break-even needs 5.85k/yr). Implicit cost is the decisive term.

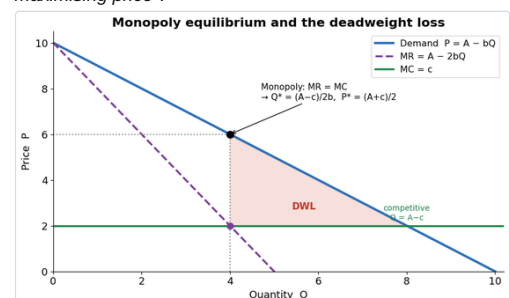
2.2 Monopoly — $MR = MC$

Single seller faces whole-market demand; sets **$MR = MC$** then reads P off demand. Linear $P = A - bQ \Rightarrow MR = A - 2bQ$ (same intercept, double slope).

► **Method / use when**

$$Q^M = \frac{A-c}{2b}, \quad P^M = \frac{A+c}{2} \quad (\text{const. } MC=c)$$

$DWL = \frac{1}{2}(P^M - MC)(Q^C - Q^M)$. Trigger: "single firm", "profit-maximising price".



Produce at $MR = MC$, price off demand; red triangle = DWL vs competitive $P = MC$.

► **Example.** $P = 1300 - 5q$, $MC = 50 + 10q$. $MR = 1300 - 10q = 50 + 10q \rightarrow q^* = 62.5$, $P^* = 987.5$.

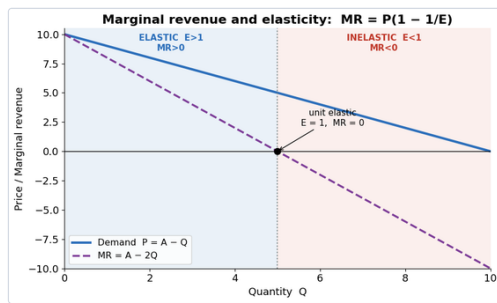
2.3 Elasticity, MR & the Lerner index

$$MR = P(1 - 1/|E|) \quad L = \frac{P - MC}{P} = 1/|E|$$

A monopolist always operates on the **elastic** part ($|E| > 1$, $MR > 0$); never the inelastic part ($MR < 0 < MC$).

► **Method / use when** Constant-elasticity pricing:

$P^* = MC / (1 - 1/|E|)$. More elastic buyers → lower markup.
Trigger: "elasticity is...", "is the firm pricing right?", "student discount".



$MR > 0$ on the elastic half, $= 0$ at midpoint, < 0 on the inelastic half.

► **Example.** $MC = 16$, $|E| = 5 \rightarrow P^* = 16 / (4/5) = 20$. Students $|E| = 9 \rightarrow P^* = 18$ (a 10% discount). $|E| = 2 \rightarrow L = 1/2$, so $P = 2 \cdot MC$.

► **Example. True/False:** $P = 100 - 10Q \rightarrow MR = 100 - 20Q = 0$ at $Q = 5$. For $Q > 5$, $MR < 0 < MC$ → a profit-maximiser **never sells > 5m units** (always elastic side) — true regardless of the cost function.

2.4 Revenue maximisation & capacity

Max revenue (not profit) where **$MR = 0$** (i.e. $|E| = 1$). A binding capacity may force a corner.

► **Method / use when** Set $MR = 0 \rightarrow Q, P$. If demand at the set price exceeds capacity, you sell out at capacity.
"Wasted" empty seats are the price of a single uniform price. Trigger: "maximise revenue", "stadium/flight capacity", "over/under-priced?".

► **Example.** $P = 40 - 0.5Q$ (k seats), cap 56k, price \$10 → demand 60k → sell 56k, $TR = 560k$. Rev-max: $MR = 40 - Q = 0 \rightarrow Q = 40k$, $P = \$20$, $TR = 800k$ → **underpriced**; optimum leaves 16k seats empty.

2.5 Taxes on a monopoly

Lump-sum / profit tax: doesn't enter $MC \rightarrow P, Q$ **unchanged** (only participation). **Per-unit tax t:** $MC \rightarrow MC + t \rightarrow P$ rises, Q falls.

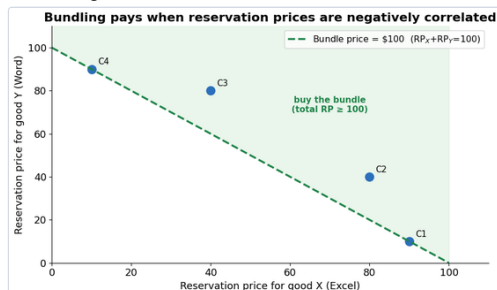
► **Method / use when** Incidence (linear demand slope b , MC slope m): **$\Delta P / \Delta t = b / (2b + m)$** . Trigger: "government imposes a tax", "who bears it?".

► **Example.** $P = 1300 - 5q$, $MC = 50 + 10q$ ($b = 5, m = 10$). Per-unit $t = 300$: $MC + t \rightarrow q = 62.5 \rightarrow 47.5$, $P = 987.5 \rightarrow 1062.5$. Pass-through $\Delta P / \Delta t = 5/20 = 25\%$. Lump-sum 5000: q, P unchanged.

2.6 Bundling

Sell goods together at one price. Profitable when buyers' reservation prices are **negatively correlated** (the bundle smooths heterogeneity).

► **Method / use when** Compare **separate** vs **pure bundle** vs **mixed** (bundle + singles @ premium / versioning). Walk candidate prices; profit = (#buyers) × (price - MC). Trigger: "sells two goods", "season ticket", "WTP table".



Negatively-correlated values cluster total WTP → one bundle price captures almost everyone.

► **Example.** Excel/Word WTPs (90,10), (80,40), (40,80), (10,90). Separate ≈ 320; pure bundle @ 100 → 400; mixed (bundle @ 120 + single @ 90) → 420.

► **Example. Season tickets** (2 seats/concert, 0 cost): separate-pricing best = 44; pure season bundle @ 23 (C1, C3 buy) = 46; versioned (full @ 26 + 2-concert pass @ 23) = 49. Mixed/versioning wins again.

2.7 Price discrimination — overview

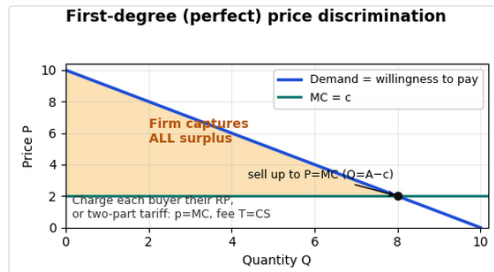
Charge different prices unrelated to cost, to capture consumer surplus. **Needs:** segments with different elasticity, price control, WTP inference, and **no arbitrage** (warranties void on resale, services, remote outlets, coupons).

► **Method / use when** Three degrees, each with its own graph & solve method below. Always start from **$MR = MC$** (per segment / per type). Trigger: "different prices", "discriminate", "separate the markets".

2.7a First-degree (perfect PD)

Firm knows each buyer's WTP → charges each their reservation price → captures the **entire** consumer surplus; quantity is efficient (last unit at $P = MC$).

► **Method / use when Solve:** sell up to the Q where demand = MC; charge each unit its demand height. In practice = a **two-part tariff**: set $p = MC$ and fixed fee $T = CS$ at that price (a separate fee per buyer when types are known).



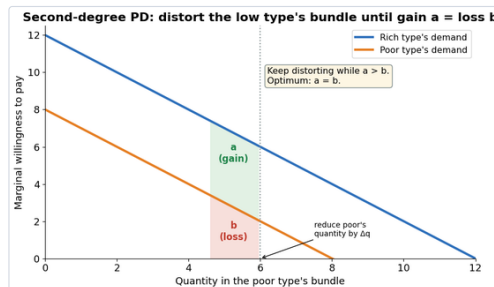
Perfect PD / two-part tariff: sell to $P = MC$ and take the whole shaded triangle via the fee.

► **Example.** $C1: q = 40 - 2P$, $C2: q = 20 - P$, $MC = 6$. Discriminating TPT: $p_1 = 6$; $T_1 = CS_1 = 1/2(20 - 6) \cdot 28 = 196$, $T_2 = 98 \rightarrow \Pi = 294$ (all surplus taken).

2.7b Second-degree (menu / screening)

Types **unobservable** → offer a **menu** (versions / quantity bundles) so buyers self-select; must satisfy **incentive compatibility** (each type prefers its own option).

► **Method / use when Solve:** high type gets $p = MC$ (efficient Q) but keeps an **information rent**; **distort the low type's quantity down** while marginal gain (a rent saved) > marginal loss b ; stop at **$a = b$** . "No distortion at the top."



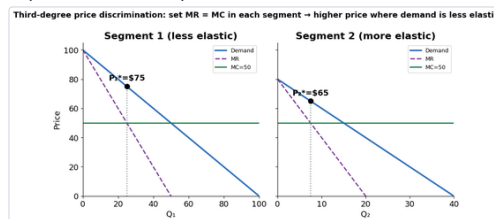
Shrink the low type's bundle while gain $a > \text{loss } b$ (no distortion at the top).

► **Example.** Same $C1/C2$, screening menu: $p_1 = MC = 6$ ($T_1 = 171.5$, low fee = info rent), $p_2 = 13 > MC$ ($T_2 = 24.5$) → $\Pi = 245$ — between uniform (220.5) and perfect PD (294).

2.7c Third-degree (group pricing)

Firm observes a **group label** (student/adult, peak/off-peak, country) and sets a separate price per group; resale between groups blocked.

► **Method / use when Solve:** in each segment set **$MR_i = MC \rightarrow Q_i, P_i$** ; the **less-elastic** group pays the **higher** price ($P_i = MC / (1 - 1/|E_i|)$). With one shared **rising** MC, MC depends on **total** output.



3rd-degree: $MR = MC$ in each segment; the less-elastic segment pays more.

► **Example.** $Q = 500 - P$, $MC = 100$. Uniform: $Q = 200, P = 300$. Split ($MR_i = MC$): business 110 @ **375**, students 90 @ **250** (total CS falls though Q is unchanged).

► **Example. Capped/government buyer + rising MC:** public $P = 100 - X$, $MC = X$. Fill govt cap 10 @ 50; then $100 - 2X_p = X_p + 10 \rightarrow X_p = 30, P_{pub} = 70, Q = 40$.

2.8 Uniform tariff & the value of information

When the firm **can't tell types apart** it must offer one uniform two-part tariff to everybody — the practical contrast with the 1st-degree case above.

► **Method / use when Solve:** a single fee can't exceed the **smaller** buyer's surplus → $T = CS_{\text{small}}(p)$; raise the per-unit p above MC to claw margin back from the big buyer; maximise total Π over p .

Value of type info = $\Pi_{\text{discriminate}} - \Pi_{\text{uniform}}$.

► **Example.** $C1: q = 40 - 2P$, $C2: q = 20 - P$, $MC = 6$. Uniform: $p^* = 9.5$, $T = 55.1 \rightarrow \Pi = 220.5$ (vs 294 if types known → info worth 73.5).

► **Example.** $q_1 = 9 - p$, $q_2 = 12 - p$, $MC = 2$. Discriminate $\Pi = 74.5$; uniform $p^* = 3.5 \rightarrow \Pi = 51.25 \rightarrow$ max to pay for type info = **23.25**.

2.9 Bertrand — price competition

Homogeneous goods, simultaneous prices, buyers go to the cheapest.

► **Method / use when Symmetric** $MC = c \rightarrow P^* = c, \Pi = 0$ (Bertrand paradox: 2 firms = perfect comp.). **Asymmetric** $c_1 < c_2$: efficient firm **limit-prices** at $\min(c_2, \text{own monopoly price})$. Entry: enter if $\Pi - F \geq 0$. Trigger: "compete in prices", "identical product".

► **Example.** $P = 100 - Q$, $MC_1 = 10$, $MC_2 = 80$. Monopoly price $55 < 80 \rightarrow$ firm 1 charges **55** ($\Pi = 2025$), firm 2 out. If $MC_2 = 50 < 55 \rightarrow$ limit-price **50**, $\Pi = 2000$.

► **Example. Cost-regime cutoff:** $Q = 10 - P$, $MC_1 = 6$, $MC_2 = c$. If $2 < c < 6$, efficient firm 2 limit-prices at 6, $\Pi_2 = 4(6 - c)$. If $c < 2$, firm 2's monopoly price $(10 + c)/2 < 6$ so it just monopolises. **Entry cost:** enter iff $\Pi - F \geq 0$ (e.g. $F \leq 2000$).

2.10 Differentiated (monopolistic) competition

Imperfect substitutes; each firm keeps some power over its own demand curve.

► **Method / use when** Demand $q_i = \alpha - \beta p_i + \gamma p_j$ (or $P_i = a - Q_i - gQ_j$). Set $\partial \Pi_i / \partial p_i = 0 \rightarrow$ best responses → intersect (use symmetry). A merger raises both prices & profits (internalises the cross-price effect).

► **Example.** $q_1 = 168 - 2P_1 + P_2$, $c = 0$. FOC $P_1 = (168 + P_2)/4 \rightarrow$ symmetric $P = 56$, $q = 112$, $\Pi = 6272$. Merger: $P = 84$, $\Pi / \text{firm} = 7056$ (both gain).

► **Example. Choosing the mode:** $p_i = 24 - q_i - 0.5q_j$, $MC = 0$. Cournot $\Pi = 92.16$; Bertrand $\Pi = 85.33$. In the 2×2 mode game, **Quantity is dominant** → NE = $(Q, Q) =$ Cournot (softer competition).

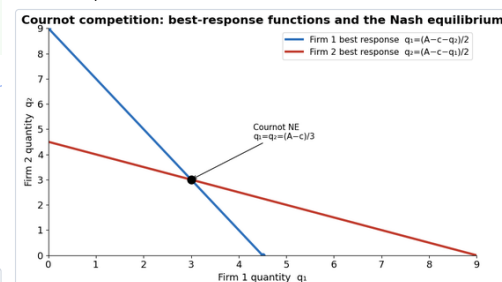
2.11 Cournot — quantity competition

Firms pick quantities simultaneously; price clears total Q . Reaction functions slope down (strategic substitutes).

► **Method / use when** FOC → BR $q_i = (A - c - q_j)/2$; intersect.

$$q_i = \frac{A - c}{3}, P = \frac{A + 2c}{3}, \Pi_i = \left(\frac{A - c}{3}\right)^2$$

N firms: $q_i = (A - c)/(N + 1)$, $P = (A + Nc)/(N + 1)$.



Downward best responses cross at the Cournot point $q = (A - c)/3$.

► **Example.** $P = 120 - Q$, $MC = 70 \rightarrow q_i = 50/3 \approx 16.7$, $P = 86.7$, $\Pi = 277.8$ each.

► **Example. Asymmetric (convex costs):** $P = 550 - Q$, $MC_1 = q_1$, $MC_2 = 2q_2$. FOCs $550 - 3q_1 - q_2 = 0$, $550 - q_1 - 4q_2 = 0 \rightarrow q_1 = 150, q_2 = 100$ (cheaper firm larger share).

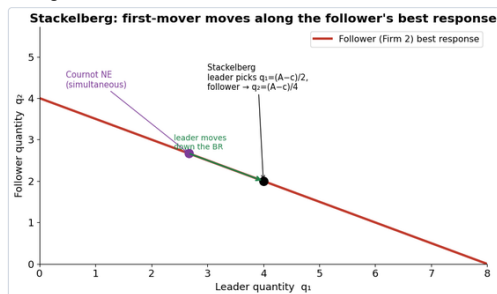
2.12 Stackberg — sequential quantity

Leader chooses q_1 first; follower best-responds. Leader substitutes the follower's BR into its own profit (**backward induction**).

► **Method / use when**

$$q_1 = \frac{A-c}{2}, q_2 = \frac{A-c}{4}, Q = \frac{3(A-c)}{4}$$

First-mover advantage (leader ↑ output & profit; follower ↓). Rising MC mutes the effect.



Leader commits to a larger quantity along the follower's BR line.

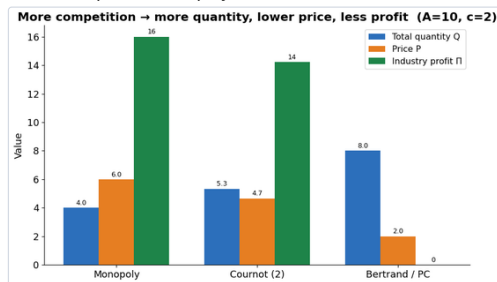
Example. $P=120-Q$, $MC=70$: leader $q_1=25$, follower $q_2=12.5 \rightarrow \Pi_1=312.5 > 277.8$ (Cournot) $> \Pi_2=156.25$.

Example. Convex costs mute it: $P=550-Q$, $MC_1=q_1$, $MC_2=2q_2$. Leader $q_1=165$ (vs Cournot 150), follower 96.25 (vs 100) — smaller shift, because each extra unit costs the leader more.

2.13 Cartel, cheating & merger

A cartel maximises **joint** profit (acts as a monopolist) then splits output — jointly best but **not** a Nash eq.: each firm gains by secretly over-producing.

► **Method / use when** Cartel: solve monopoly Q , split. Cheat: best-respond to partner's cartel output. Merger price band: **max buyer pays** = Π^{monopoly} — **own duopoly** Π ; min seller accepts = its duopoly Π .



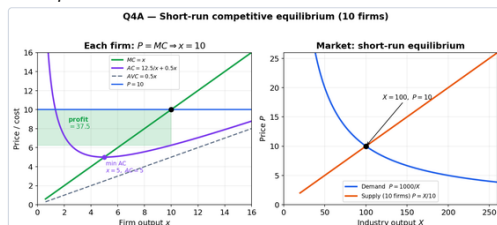
Monopoly → Cournot → Bertrand: $Q \uparrow$, $P \downarrow$, industry $\Pi \downarrow$.

Example. $P=200-Q$, $MC=80$. Cournot $\Pi=1600$ ea. Cartel $q=30$, $\Pi=1800$. Cheat $q=45$, $\Pi=2025$ (cheat > cartel > Cournot). Monopoly $\Pi=3600 \rightarrow$ buyer pays $\leq 3600-1600=2000$.

2.14 Perfect competition (SR & LR)

Price takers: each sets $P=MC$. Short run: profit can be $\pm$; produce while $P \geq \min AVC$ (**shutdown**). Long run: free entry $\rightarrow P=\Pi=0$, $P=\min AC$, each firm at efficient scale; demand sets only the **number of firms**.

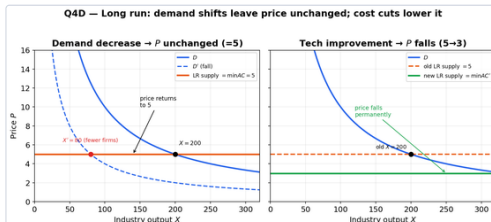
► **Method / use when** SR: firm supply $x(P)=MC^{-1}$; sum to X ; clear vs demand. LR: $P=\min AC$ (where $MC=AC$). **Co-existing activities** → equate net returns (gross gap = cost difference). Trigger: "many firms / free entry", "long run", "two options coexist".



Left: firm at $P=MC$, profit rectangle. Right: market clears where supply meets demand.

Example. $P=1000/X$, $C=12.5+0.5x^2$ ($MC=x$), 10 firms. SR: $x=P$, $X=10P$, $P=1000/(10P) \rightarrow P=10$, $x=10$, $\Pi=37.5$. LR: $\min AC$ at $x=5$ ($=5$) $\rightarrow P=5$, $X=200$, **40 firms**. Demand $\downarrow \Rightarrow$ LR price still 5, fewer firms.

Example. Arbitrage / coexistence: Airbnb (furnished, 10k/yr furniture) vs long-let coexist $\Rightarrow$ net returns equal: $R_{\text{Airbnb}} - 10k = R_{\text{long}} \rightarrow$ gross gap = **10k** (a compensating differential).



Long run: a **demand** fall leaves $P=\min AC$ unchanged (only fewer firms); a **cost** cut lowers the long-run price.

2.15 Vertical relations — double marginalisation

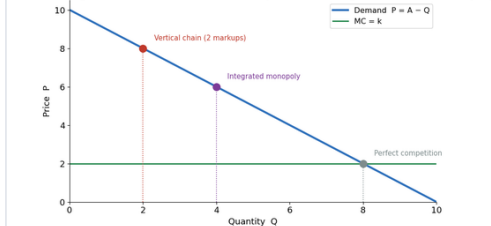
Producer ($MC=k$) sells wholesale w to a retailer who then sets P . Two stacked markups $\rightarrow$ higher P , lower Q than one integrated monopolist — worse for everyone.

► **Method / use when** Backward induction: retailer sets Q given w ; producer then sets w .

$$Q^{VR} = \frac{A-k}{4}, w = \frac{A+k}{2}, P = \frac{3A+k}{4}$$

Fix: **two-part tariff** ($w=MC$ + fee), vertical merger, or RPM $\rightarrow$ restores integrated outcome.

Double marginalisation: the vertical chain prices even higher than one monopoly



The chain stacks two markups $\rightarrow$ even higher P / lower Q than one integrated monopoly.

Example. $P=200-Q$, $MC=0$. Integrated: $Q=100$, $P=100$, $\Pi=10000$. Chain: $w=100$, $P=150$, $Q=50 \rightarrow \Pi=5000+2500=7500 < 10000$. TPT ($w=0$ +fee) restores 10000.

LECTURE 3 — GAME THEORY

3.1 Dominance & IESDS

Normal form: rows= P_1 , cols= P_2 , cell= (P_1, P_2) . **Dominant** = best vs every rival action; **dominated** = never played.

► **Method / use when** Strict dominance: $>$ in every column. Weak: $\geq$ all, $>$ one (say which you use). **IESDS**: delete strictly dominated strategies repeatedly (order-free; never removes a NE).

Example. Hire vs Not $(5,5)/(3,6)/(6,3)/(4,4)$: Hire beats Not in both columns $\rightarrow$ Hire dominant $\rightarrow$ NE (Hire, Hire) = $(4,4)$, worse than $(5,5)$ — Prisoner's Dilemma.

Example. IESDS: delete any strictly dominated row/column, then re-scan the smaller matrix. Order doesn't matter and a NE is never deleted — so if one cell survives, it is the unique NE.

3.2 Pure Nash & parametric conditions

Nash eq. = no player gains by a unilateral deviation (mutual best responses).

► **Method / use when** Underline method: mark P_1 's best in each column & P_2 's best in each row; doubly-marked cell = pure NE. For **parametric** games, write each profile's two BR inequalities; for a **unique** NE, also negate every other profile's conditions. Trigger: "for which values of $a, b, \dots$ is it a NE".

Example. $U: (3,2)/(1,4) / D: (2,1)/(4,3)$. Col $L \rightarrow U$, col $R \rightarrow D$; row $U \rightarrow R$, row $D \rightarrow R \rightarrow$ **NE = (D, R) = (4, 3)**.

Example. Parametric: rows U/D , cols R/L , payoffs $U: (c, 1)/(1, a)$ $D: (d, 1)/(2, b)$. (Right, Down) is the **unique** NE iff $b < 1$, $d \geq c$, and $(d < c \text{ or } a > 1)$ — the last clause kills (Right, Up); $b < 1$ kills (Left, Down).

3.3 The game zoo

Recognise the structure to know how many equilibria to expect:

► **Method / use when**

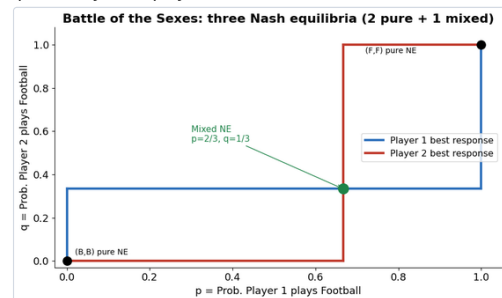
Game	#NE	Key
Prisoner's Dil.	1	dominant $\rightarrow$ Pareto-inferior
Battle of Sexes	3	2 pure + 1 mixed
Coordination / network	2	Pareto- vs risk-dominant; lock-in
Matching pennies / odd-even	1	0 pure, mixed $(\frac{1}{2}, \frac{1}{2})$
Rock-Paper-Sc.	1	0 pure, mixed $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Example. Network adoption: payoffs add bonus c if both pick the same model. (A, A) is a NE iff $c \geq b$ (the matcher's bonus beats its preference loss); if $a=b < c$ there are two NE (A, A) & (B, B) .

3.4 Mixed-strategy NE

When no pure NE exists (or for extra equilibria), players randomise.

► **Method / use when** Make the **opponent indifferent**: set their EU of each pure strategy equal, solve for *your* mixing probability. Trigger: "no pure NE", "how many equilibria", "probability each plays...".



BoS best-response correspondences cross 3×2 pure corners + 1 interior mixed NE.

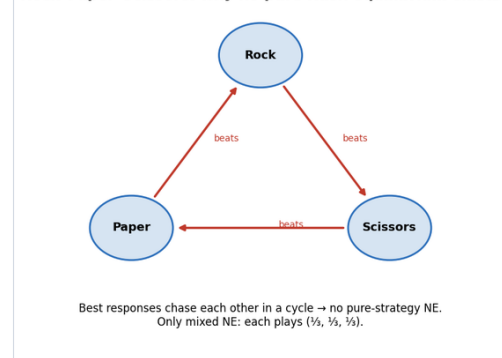
Example. $U: (0,2)/(3,0) / D: (2,0)/(0,3)$: $3-3q=2q \Rightarrow q=\frac{1}{3}$; symmetry $p=\frac{1}{3}$. Battle of Sexes $(2,1)$: $p=\frac{2}{3}$, $q=\frac{1}{3}$.

3.5 Iterated dominance unravelling

Even cooperative-looking outcomes collapse if undercutting is always individually better.

► **Method / use when** Spot the marginal incentive to deviate by ϵ and iterate to the floor. Same lesson as PD: rational $\neq$ efficient. **Escape**: repeated play (tit-for-tat / folk theorem), binding contracts, social preferences.

Rock-Paper-Scissors: why no pure Nash equilibrium exists



Best responses cycle $\rightarrow$ no pure NE; only the uniform mixed NE.

Example. Traveller's dilemma, report 180–300, lower report +5 bonus, higher –5. (300,300) is Pareto-best but each undercuts $\rightarrow$ unravels to unique NE **(180,180)**.

LECTURE 4 — PRICE COMPETITION WITH COMPLEMENTARY GOODS

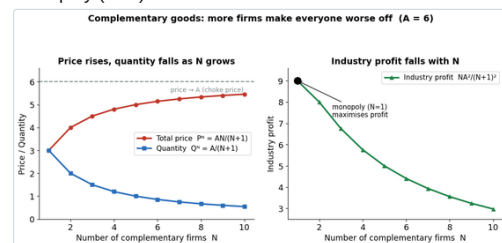
4.1 Perfect complements & N-firm pricing

Buyer needs ALL components $\rightarrow$ cares only about the **total** price $P=\sum p_i$; demand $Q=A-P$. Each firm prices its piece taking the others' as given.

► **Method / use when** FOC firm i : $A-P-p_i=0$; impose symmetry $p_i=P/N$.

$$P^N = \frac{AN}{N+1}, p_i = Q = \frac{A}{N+1}, \Pi_{\text{tot}} = \frac{NA^2}{(N+1)^2}$$

Monopoly ($N=1$): $P=A/2$.



More complementary firms: total price $\rightarrow$ choke price; industry profit maximised at $N=1$.

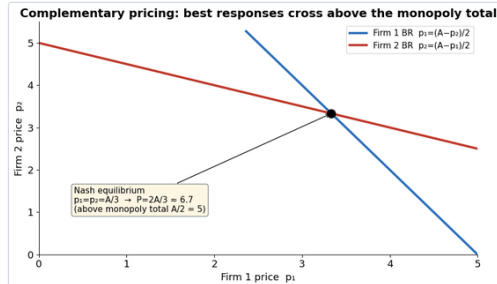
Example. $A=10$, $N=2$ (toll road): $p_i=10/3$, $P=20/3=6.67$, $Q=10/3$, $\Pi_i=100/9$. vs monopoly $P=5$, $\Pi=25 \rightarrow$ firms jointly earn less.

Example. Adding a firm hurts: $A=12$. $N=1$: $P=6$, $\Pi=36$. $N=3$: $P=AN/(N+1)=9$, $Q=3$, $\Pi_{\text{tot}}=27 < 36$. Each extra complement raises P and shrinks the pie — vertical integration reverses it.

4.2 Double marginalisation / pricing externality

Each firm's markup raises the total price and shrinks demand for **all** firms, but it ignores that loss — a **negative pricing externality**.

► **Method / use when** So more complementary firms → higher P, lower Q, lower Π — the exact opposite of substitutes. As $N \rightarrow \infty$, $P \rightarrow A$, $Q \rightarrow 0$.



Price best responses cross above the monopoly total — the externality, drawn.

► **Example.** $N: P = AN/(N+1)$ rises in N ($A/2$, $2A/3$, $3A/4$, ...). Industry profit falls monotonically from the monopoly value.

4.3 Fixes & substitutes vs complements

Internalise the externality: **vertical integration** (→ $P = A/2$), **two-part tariff**, or **revenue sharing**.

► **Method / use when** Read the cross-price sign first. Substitutes (compete) → more firms lower P, raise welfare. Complements → more firms raise P, lower welfare. Trigger: "toll booths / console+game / supply chain".

► **Example.** Two differentiated substitutes $q_i = A - p_i + \gamma p_j$: more competition → output > independent benchmark → welfare ↑. Complements: welfare ↓.

Σ FORMULA BANK (LINEAR DEMAND, $MC=C$)

Structure	Q	P
Monopoly	$(A-c)/2b$	$(A+c)/2$
Cournot (2)	$2(A-c)/3b$	$(A+2c)/3$
Cournot (N)	$N(A-c)/(N+1)b$	$(A+Nc)/(N+1)$
Stackelberg	$3(A-c)/4b$	$(A+3c)/4$
Bertrand / PC	$(A-c)/b$	c
Vert. chain	$(A-k)/4$	$(3A+k)/4$
Complements (N)	$A/(N+1)$	$AN/(N+1)$

$MR = A - 2bQ$ · $MR = P(1-1/|E|)$ · $L = (P-MC)/P = 1/|E|$ · $P^* = MC/(1-1/|E|)$ · $CS = \frac{1}{2}(P_{\max} - p)Q$ · tax pass-through $b/(2b+m)$ · TPT: $p = MC$, $T = CS$.

► EXAM REFLEXES — READ THE TRIGGER WORDS

Asymmetric info & pricing

- "Which types trade / who buys" → test pools for self-consistency (REE); top type exits first.
- "Uninsured" → $P = MC$. "Fully insured" → $P = 0$, over-consume; fair premium on the insured quantities.
- "Sold separately / single price" → walk the WTP ladder; $\Pi = n \cdot (p - MC)$.
- "Membership card / two-part tariff" → $p = MC$, fee = CS; never price the good at 0 if MC rises.
- "Value of identifying types / info" → $\Pi_{\text{discriminate}} - \Pi_{\text{uniform}}$ (a difference, not a level).
- "Government / capped buyer + rising MC" → fill the priced channel to its cap; $MR = MC$ on total Q.
- "Per-unit vs lump-sum tax" → per-unit shifts MC (incidence $b/(2b+m)$); lump-sum/profit leaves P, Q unchanged.

Competition & market structure

- **Asymmetric Bertrand** → efficient firm limit-prices at min(rival MC, own monopoly price).
- "Sells through a distributor" → backward induction; $\Pi_{\text{chain}} < \Pi_{\text{integrated}}$; fix with TPT/merger.
- "Chooses first" → Stackelberg (sub follower BR first). "Simultaneous quantities" → Cournot.
- "Max price to acquire a rival" = monopoly Π – buyer's duopoly Π ; min = seller's duopoly Π .
- "Long-run, free entry" → $P = \min AC$, $\Pi = 0$; demand sets only the number of firms, not the LR price.
- "Two activities coexist" → equate **net** returns; gross-return gap = cost difference.
- **Differentiated mode choice** → Quantity dominates Price → firms land at Cournot.

Game theory & complements

- "Odd/even / matching pennies" → no pure NE, unique mixed NE ("no NE" is the trap).
- **Battle of the Sexes** → 3 equilibria (2 pure + 1 mixed); **RPS** → uniform mixed.
- "For which values of...is it a NE" → write each profile's BR inequalities; for uniqueness negate the others.
- **Mixed NE** → make the **opponent** indifferent; solve for your own probability.
- **Substitutes** compete → $P \downarrow$, welfare ↑. **Complements** → double marginalisation, $P \uparrow$, welfare ↓ — check the cross-price sign.
- "Toll booths / console+game / supply chain" → complements: more firms → higher total price.

Coffee carts — a whole-market sweep

4 carts, each $TC = q^2 + 64$ ($MC = 2q$); demand $Q = 440 - 20p$.

(a) **Perfect competition.** $P = MC = 2q \rightarrow q = p/2$; industry supply $Q = 2p$. Clear $2p = 440 - 20p \rightarrow p = 20$, $Q = 40$, each cart $q = 10$.

(b) **Merge → monopoly + coffee card (TPT).** $TC = 0.25Q^2$ ($MC = 0.5Q$). Optimal TPT: per-unit $p = MC$, so $22 - Q/20 = 0.5Q \rightarrow Q = 40$, $p = 20$; fixed fee $T = CS = \frac{1}{2}(22 - 20) \cdot 40 = 40$.

(c) **One uniform price.** MBA $Q = 220 - 10p$, others $Q = 220 - 20p$. Serving MBAs only is best → $MR = MC \rightarrow p = 132/7 = 18.9$.

(d) **PD across days.** Separable markets → $MR_i = MC$ each: MBA $p = 18.9$, others $p = 10.1$ (less-elastic MBAs pay more).

Two-part tariffs to two consumers

C1 $p = 200 - q$, C2 $p = 150 - q$, $MC = 20$.

(a) **Pay to discriminate (1st-deg).** Each $p = MC = 20$; $T_i = CS_i$; $T_1 = \frac{1}{2}(180)^2 = 16200$, $T_2 = \frac{1}{2}(130)^2 = 8450 \rightarrow \Pi = 24650$.

(b) **Uniform tariff.** Fee pinned by smaller C2: $T = \frac{1}{2}(150 - p)^2$. Maximise $\Rightarrow p^* = 45$, $T = 5512.5 \rightarrow \Pi = 17525$. Max to pay for type info = $24650 - 17525 = 7125$.

(c) **Serve one or both?** Compare serving-both profit with serving only C1 (16200). Drop the weak consumer when it only drags the shared fee down.